MIT Art, Design and Technology University's
MIT School of Computing, Pune
Department of Applied Science & Humanities

First Year B. Tech. (All Branches)
23ASH1107: Mathematical Foundation for Computing -II
UNIT – I: Linear Algebra-I
A.Y.2025-26, Semester-II

Tutorial No.:1

(Rank of Matrix, Echelon form and Normal form)

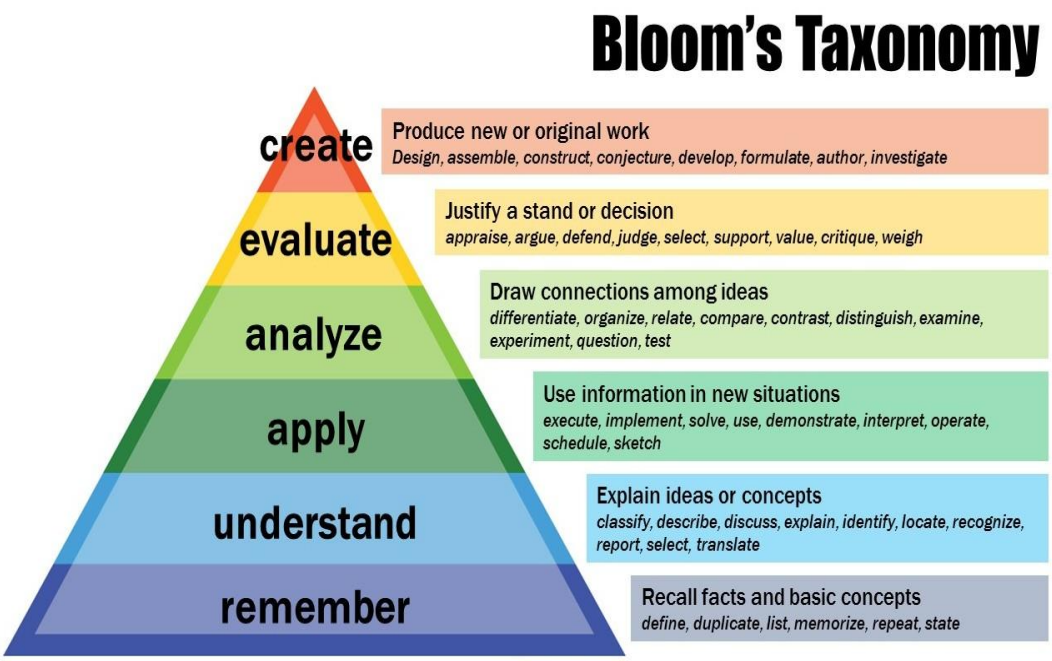
Que. No.	Question Statement	CO	POs	Blooms Level	Marks
1	a) Define rank of a matrix. b) What is the relationship between the rank of a matrix and linear independence of its rows?	CO1	PO1 PO2 PO3 PO4 PO12	1	2
2	Define Echelon form and Normal Form of a matrix. How to obtain the rank of a matrix from these forms.	CO1		1	2
3	Reduce the following matrix into Echelon form and hence find its rank $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$ **[Ans:$\rho(A) = 3$]**	CO1		2	4
4	Reduce the following matrix into Echelon form and hence find its rank $\begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 7 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ **[Ans:$\rho(A) = 4$]**	CO1		2	4
5	Reduce the following matrix to its normal form and hence find its rank $\begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ 2 & 3 & 10 \end{bmatrix}$ **[Ans:$\rho(A) = 2$]**	CO1		2	4
6	Reduce the following matrix to its normal form and hence find its rank $\begin{bmatrix} 1 & -1 & 2 & 3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 1 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$ **[Ans:$\rho(A) = 4$]**	CO1		2	4

Tutorial No.:2 *(Homogeneous and Non-homogeneous System of Linear Equations)*					
Que. No.	**Question Statement**	**CO**	**POs**	**Blooms Level**	**Marks**
1	Discuss the conditions under which a system of linear equations has (a) No solution (b) A unique solution (c) Infinitely many solutions. Also explain geometric interpretation of these cases of solutions.	CO1	PO1 PO2 PO3 PO4 PO12	1	2
2	Define a) Solution of a system of linear equations b) Consistent System and inconsistent system of linear equations	CO1		1	2
3	Define a) Homogeneous System and Non-Homogeneous System of linear equations b) Augmented Matrix b) Free Variables (Parameters)	CO1		1	2
4	Test for consistency and if consistent, solve the following system of equations: $3x + y + 2z = 3; 2x - 3y - z = -3; x + 2y + z = 4$ **[Ans:** Unique solution $x = 1, y = 2, z = -1$**]**	CO1		3	4
5	Test for consistency and if consistent, solve the following system of equations: $4x - 2y + 6z = 8; x + y - 3z = -1; 15x - 3y + 9z = 21$ **[Ans:** Infinite solution $x = 1, y = 3t - 2, z = t$**]**	CO1		3	4
6	By considering the ranks of relevant matrices examine for consistency the system of equations $x_1 + x_2 + 2x_3 + x_4 = 5,$ $2x_1 + 3x_2 - x_3 - 2x_4 = 2,$ $4x_1 + 5x_2 + 3x_3 = 7$ Solve if consistent. **[Ans:** No solution**]**	CO1		3	4
7	Investigate for what values of λ and μ the system of simultaneous equations, $x + y + z = 6;\ x + 2y + 3z = 10;\ x + 2y + \lambda z = \mu$ have (i) no solution (ii) an infinite Solution (iii) a unique solution.	CO1		3	4

	[Ans: (i) No Solution $\lambda = 3, \mu \neq 10$ (ii) Infinite Solution $\lambda = 3, \mu = 10$ (iii) Unique Solution $\lambda = 3, \mu$ can have any value. **]**				
8	Examine for non-trivial solution for the following set of equations and solve them, if so $2x_1 - 2x_2 + 5x_3 + 3x_4 = 0;$ $4x_1 - x_2 + x_3 + x_4 = 0;$ $3x_1 - 2x_2 + 3x_3 + 4x_4 = 0;$ $x_1 - 3x_2 + 7x_3 + 6x_4 = 0.$ **[Ans**: Non-trivial Solution $x_1 = \frac{5t}{9}, x_2 = 4t, x_3 = \frac{7t}{9}, x_4 = t$]	CO1	PO1 PO2 PO3 PO4 PO12	3	4
9	Show that the system of equations, $x_1 + 2x_2 + 3x_3 = \lambda x_1;$ $3x_1 + x_2 + 2x_3 = \lambda x_2;$ $2x_1 + 3x_2 + x_3 = \lambda x_3$ can possess a non-trivial solution only if $\lambda = 6$. Obtain general solution for real values of λ. **[Ans**: For, $\lambda = 6, x_1 = x_2 = x_3 = t$ **]**	CO1		3	4

Tutorial No.:3 *(Linear Transformation and Orthogonal Transformation)*					
Que. No.	**Question Statement**	**CO**	**POs**	**Blooms Level**	**Marks**
1	Define a) Linear Transformation b) Regular Transformation	CO1	PO1 PO2 PO3 PO4 PO12	1	2
2	Define: Composite Linear Transformation	CO1		1	2
3	What is the Determinant and inverse of an orthogonal matrix	CO1		1	2
4	Define a) Orthogonal Matrix **b)** Orthogonal Transformation	CO1		1	2
5	Given the transformation $Y = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Find the coordinates (x_1, x_2, x_3) in X corresponding to $(1,2,-1)$ in Y. where $Y = [y_1, y_2, y_3]^T$ **[Ans**: $(x_1, x_2, x_3) = (-1, 3, 0)$ **]**	CO1		3	4
6	Express each of the transformations $x_1 = 3y_1 + 5y_2;\ \ x_2 = -y_1 + 7y_2;$ $y_1 = z_1 + 3z_2;\ y_2 = 4z_1$ in the matrix form and find the composite transformation which express x_1, x_2 in terms of z_1, z_2. **[Ans:** $x_1 = 23z_1 + 9z_2, x_2 = 27z_1 - 3z_2$]	CO1		3	4
7	Check whether the following matrix is orthogonal or not? $\frac{1}{13}\begin{bmatrix} -12 & -5 \\ 5 & -12 \end{bmatrix}$ **[Ans:** Yes]	CO1		2	2
8	Check whether the following matrix is orthogonal or not? $\begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & \frac{-1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{bmatrix}$	CO1		3	4

	[Ans: Yes]				
9	Determine the value of a, b, c when the following matrix is orthogonal $\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & a \\ \frac{2}{3} & \frac{1}{3} & b \\ \frac{2}{3} & \frac{-2}{3} & c \end{bmatrix}$ **[Ans:** $a = \pm\frac{2}{3}, b = \pm\frac{2}{3}, c = \pm\frac{1}{3}$ **]**	CO1	PO1 PO2 PO3 PO4 PO12	3	4

looms level no	Blooms Taxonomy terms
6	
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